Vishal Chaudhary

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Education

TU Dortmund University — Dortmund, Germany
M.Sc. in Automation and Robotics — *Oct 2021 – Dec 2025*

- **Master's Thesis** (Nov 2025): *"Definition and Development of a Low-Voltage Grid Digital Twin using the Asset Administration Shell as a Single-Source-of-Truth for Power System Data and Services"* – designed and implemented an AAS-centric (IEC 63278) Digital Twin unifying CIM/CGMES planning data, asset datasheets, and real-time HIL telemetry (Modbus/DLMS) under a hierarchy of Asset Administration Shells keyed by CIM-derived global asset IDs
- Built a closed-loop Distribution System State Estimation (DSSE) service (PandaPower, Weighted Least Squares) that consumes its inputs entirely from the AAS and writes estimated states back into it (also broadcasting to InfluxDB and RabbitMQ for downstream consumption); validated on a 15-bus/13-line network with 79 input measurements, converging in ~611ms per cycle with Maximum Absolute Error ~0.001 p.u. and Mean Absolute Percentage Error <0.1% against measured voltages
- First-authoring a paper based on these findings for conference/journal submission in 2026
- **Master Group Project** – *Transparent Energy Transition: Generation of grid models for the digitization of "blind" low-voltage grids*: integrated group-generated GIS XML data with real-time streams from a Typhoon HIL model to engineer compliant CIM CGMES grid models, enabling precise power-flow analysis
- **Academic Project** – *Unsupervised Learning of Parametric Optimization Problems*: showed a neural network trained via a custom loss function can solve parametric optimization problems to the same precision as supervised methods

GLA University — Uttar Pradesh, India
B.Tech. in Mechanical Engineering — *Aug 2015 – Jun 2019*

- Focus: Control Theory, Embedded Systems
- **Bachelor's Project**: *Design and Implementation of a PID-based Flight Control System for 6-DOF Quadcopter Stability* – developed an embedded C++/Arduino controller with tuned PID loops achieving consistent autonomous takeoff and hover

Experience

Research Assistant — Aug 2023 – Mar 2026
IE3, TU Dortmund

Smart-Meter Data Integration (SCD / IEC 61850)

* Built a dual-protocol smart-meter/substation data-acquisition service driven by SCD (IEC 61850 SCL) configuration files, polling Modbus TCP registers from IEDs and running TLS-secured DLMS/COSEM listeners in parallel, both mapped into a common IEC 61850 logical-node (MMXU/TCTR) model
* Implemented SCD parsing (IEC 61850 SCL XML) into Modbus register and DLMS/COSEM connection mappings, including word-order- and datatype-aware register decoding (IEEE floats, 16/32-bit values) for dynamic per-IED Modbus TCP clients
* Implemented TLS-secured DLMS/COSEM listeners (one process per SMGW port) decoding CMS-encrypted smart-meter messages and parsing TAF10 payloads into IEC 61850 logical nodes
* Built the InfluxDB time-series storage layer for the resulting measurement data (Grafana-ready), and the multiprocessing orchestration running Modbus polling and DLMS listening concurrently on a configurable interval

Typhoon HIL Grid Model Automation

* Designed and built an end-to-end microservices pipeline (FastAPI gateway plus async, job-based model creation/compilation/execution services, each Dockerized) automating the full Typhoon HIL workflow from grid data to physical hardware execution, replacing manual model construction
* Implemented the compilation and execution services: compiling models via Typhoon's SchematicAPI and running validated models on physical HIL606 devices addressed by serial number, passing model artifacts between pipeline stages through MinIO S3
* Designed and implemented a budget- and length-aware TLM coupling-placement algorithm deciding, per grid line, between core, device, or no coupling – replacing a heuristic that over-cut dense grids – reducing total TLM boundaries on a reference grid from 13 to 8 while eliminating cuts on electrically-unsuitable short lines

- Modeled two independent hardware constraints (SP-sources-per-processing-core and SP-subcircuits-per-device) with separate tunable budgets, and built a capacity-aware best-fit algorithm to pack grid components across multiple parallel HIL devices, splitting oversized islands automatically
- Built the model-creation pipeline (topology, bus/load creation, external grid/slack bus, Modbus/TCP communication setup, meter/switch placement) driving the Typhoon HIL SchematicAPI

CIM-CGMES Domain Pipeline

- Designed and containerized a microservices pipeline that converts GIS power-grid data into standards-compliant CIM CGMES models, deployed across 8 services with Podman Compose over a two-network topology (internal service mesh + gateway-only external network)
- Built an API gateway (FastAPI) implementing prefix-based reverse-proxy routing to internal services, with unified health-check and API-docs-index endpoints
- Built a pipeline orchestrator coordinating the end-to-end conversion flow (topology generation, anomaly detection/rectification, grid-model generation, CGMES generation) via REST, exposing a single end-to-end pipeline endpoint
- Built a shared S3/MinIO storage interface service (upload, download, list, delete, folder-zip) used by every stage of the pipeline for intermediate data exchange
- Implemented the CGMES generation service, converting grid-model data into 6 CIM XML profiles (EQ, TP, SSH, SV, DL, GL) per transformer, including schema-normalization and data-robustness fixes

cim_cgmes_classes (Python library)

- Authored and published a standalone Python library (private PyPI) providing an object-oriented model of 27 CIM/CGMES classes (transformers, switches, lines, loads, terminals, connectivity/topological nodes, and SSH/SV/GL profile containers), used as the shared domain model consumed by the CGMES generation service
- Implemented RDF/XML serialization (namespaced element construction, UUID-based mRID identity) so any equipment class can emit standards-compliant CIM CGMES XML
- Modeled detailed power transformer semantics: substation/voltage-level hierarchy, winding connections and tap changers, and external network injection with configurable regulating control
- Extended the library with Asset Administration Shell (AAS) type mappings for BaSyx-based digital twin shells, integrating the CIM data model into the broader LVTwinShell digital-twin platform
- Set up automated testing (pytest), linting, and CI/CD (shared GitLab CI templates, pre-commit hooks) for the library's release pipeline

Technical Skills

Programming Languages: Python, C/C++
Frameworks & Libraries: FastAPI, Uvicorn, pytest, PandaPower
Infrastructure: Docker, Podman, Podman Compose, MinIO (S3-compatible object storage)
Architecture: Microservices, REST API design, reverse-proxy/API gateways, async job-based orchestration (FastAPI BackgroundTasks), service orchestration
Domain: CIM/CGMES power-system data modeling, IEC 61850 substation automation, Typhoon HIL real-time simulation, Asset Administration Shell (AAS/IEC 63278) digital twins, Weighted Least Squares state estimation
Algorithms & Optimization: Graph traversal and budget-constrained partitioning/bin-packing for hardware-aware model placement; Gurobi
Protocols & Security: Modbus/TCP, DLMS/COSEM, TLS (X.509 certs/keys)
Data & Messaging: InfluxDB (time-series), RabbitMQ, Grafana, multiprocessing-based concurrent data acquisition
Tools: Git, GitLab CI, pre-commit, setuptools/PyPI packaging
Spoken Languages: English (C1), German (B1), Hindi (Native)

Publications

- **V. Chaudhary** et al., "A Low Voltage Grid Digital Twin using the Asset Administration Shell as a Single Source of Truth," *in preparation, 2026*

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Dortmund, July 5, 2026 Vishal Chaudhary